

ANANYASREE S

+91-9844550847 • sreeananya33@gmail.com • Bangalore • [LinkedIn](#) • [GitHub](#)

Engineer with 3+ years of experience debugging enterprise-scale distributed systems at SAP. Has hands-on experience building Generative AI applications spanning RAG, LLM inference, voice AI, semantic search, tool calling and LLM evaluation. Strong in Python, FastAPI, Redis, vector databases, distributed systems, and scalable backend services, with experience taking AI systems from architecture and experimentation through evaluation and deployment.

WORK EXPERIENCE

Associate Engineer - SAP Analytics Cloud, SAP

July 2023 - Present

- **Built and enhanced** backend services for an **internal analytics platform(SAC)**, improving system usability and **reducing manual configuration effort by 40% for 300+ users**
- **Owned end-to-end resolution of critical production failures** in distributed APIs and data pipelines **serving 2000+ users**, drove systematic root-cause analysis using Splunk, network traces, and observability tooling to identify API, caching, and distributed system failures, reducing recurring incidents by 30%.
- Partnered with enterprise customers and cross-functional teams to deploy AI/ML integration solutions for production issue triage and incident summarization, directly reducing operational overhead for 2000+ users.
- Mentored junior engineers; documented debugging workflows, incident response practices, and onboarding checklists for production-scale distributed systems.

Intern, SAP

Feb 2023 - July 2023

- Triageed and resolved backend issues in production-scale distributed systems, contributing to root-cause analysis across large-scale APIs, data pipelines, and observability tooling

PROJECTS

AI Production Debugging Copilot | Python, FastAPI, Redis, FAISS, Ollama (LLMs), Transformers, RAGAS

- Designed and built an AI-powered debugging assistant for distributed systems, analyzing logs, metrics, and traces across 50+ simulated failure scenarios to assist with root-cause analysis using semantic search over a FAISS vector database.
- Implemented RAG pipeline using FAISS vector database and local LLM deployment via Ollama, enabling fully offline semantic search with inference latency under 10ms and end-to-end retrieval latency under 1s on CPU.
- Implemented model evaluation & retrieval metrics with 20+ golden dataset and 4 different validation metrics using RAGAs; benchmarked performance across query sets by iterating prompts and chunking strategy

SupportFlow AI: Agentic Real-Time Voice Sales Bot | Python, FastAPI, Faster-Whisper, Ollama/Gemma, Web Audio API

- Built an end-to-end real-time voice sales agent using Faster-Whisper, FastAPI, Ollama/Gemma, and streaming TTS, enabling multi-turn conversational lead discovery through natural speech.
- Implemented autonomous tool-calling so the LLM independently decides when lead qualification is complete and invokes a function to score and persist the lead to a CRM, without human-in-the-loop triggering.
- Instrumented every conversational turn with granular latency metrics (STT, LLM time-to-first-token, first-audio, end-to-end total) while keeping the tool-calling decision step off the critical speech-latency path.

Distributed Rate Limiter | Python, FastAPI, Redis, Prometheus

- Designed and implemented a distributed API rate-limiting service with high availability using Token Bucket and Sliding Window algorithms for concurrent requests, used Prometheus metrics to log failed requests.
- Integrated as FastAPI middleware adding under 2ms average overhead at ~1,000 req/sec sustained throughput; used Redis atomic Lua scripts for concurrency correctness across hundreds of parallel requests.
- Deployed the service on AWS ECS, validating correctness and throughput under a real multi-container, networked environment rather than a single local process.

SKILLS & PROFICIENCIES

- **Languages & Core CS:** Python, SQL, Data Structures & Algorithms, OOP, DBMS, System Design
- **Backend & Distributed Systems:** FastAPI, REST APIs, Redis, Distributed Systems, Concurrency, Scalability, Performance Optimization, Fault tolerance
- **Generative AI :** LLMs, RAG, LLM Inference & Deployment, Vector Databases, Embeddings, Semantic Search, Prompt Engineering, LLM Evaluation, Voice AI, Speech-to-Text, Text-to-Speech, LangChain, tool calling
- **Infrastructure & Tools:** AWS (EC2, CloudWatch), Docker, MySQL, FAISS, HuggingFace, Ollama, Prometheus, Splunk, Git, Linux

EDUCATIONAL BACKGROUND

BMS Institute of Technology and Management, Bangalore

Aug 2019 – May 2023

Bachelor of Technology (B.Tech) - Computer Science and Engineering, CGPA 9.10

CERTIFICATIONS

- **Generative AI Developer Certification - SAP**